Agenda

- Classification
- Logistic Review
- LDA

0. Classification

$Y \in \{1, 2, \ldots, K\}$ 0-1 Loss

$E[I(Y \neq f(X))]$

$I(\text{true}) = 1$
$I(\text{false}) = 0$

minimized by $f(x) = \underset{k}{\operatorname{argmax}} P(Y = k \mid X = x)$

$=$ ↑

Bayes classifier (unknown)

Approximate it!

1. Logistic Regression

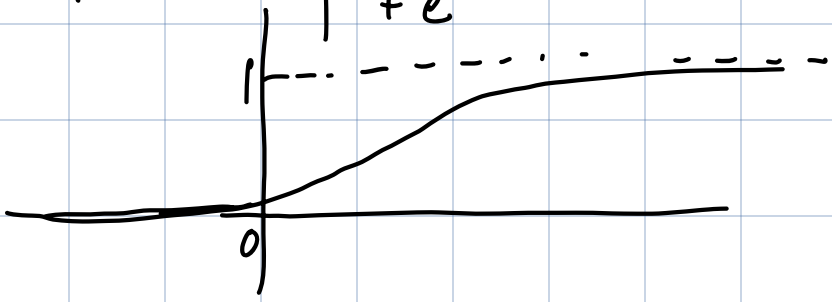
$Y \in \{1, 2\}$

Assumption: log odds is linear

$$\log \frac{p(x)}{1 - p(x)} = \beta_0 + x^T \beta$$

1.1 $p(x) = P(Y=2 \mid X=x)$

$$= \frac{e^{\beta_0 + x^T\beta}}{1 + e^{\beta_0 + x^T\beta}} = \frac{1}{1 + e^{-(\beta_0 + x^T\beta)}}$$



Find $\hat{\beta}, \hat{\beta}_0$ using MLE, but no closed form solution. Use gradient descent, Newton-Raphson

1.2 Predictions: $\hat{p}(x) = \frac{1}{1 + e^{-(\hat{\beta}_0 + x^T\hat{\beta})}} > c$, $Y=2$

else $Y=1$

$c = 0.5$ natural choice, not usually best

$\hat{p}(x) = (.1, .6, .3, .4, .7, .9, .8)$

predicted class $= (0, 1, 0, 0, 1, 1, 1)$

actual $= (0, 1, 0, 1, 0, 1, 1)$

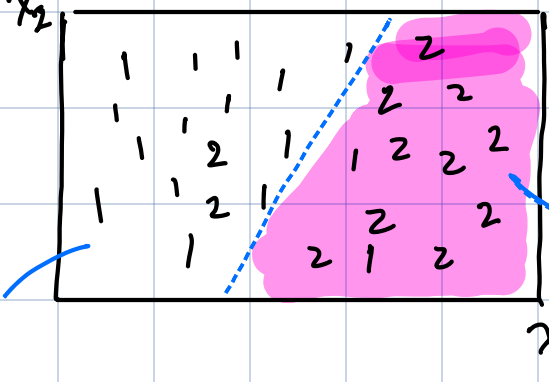
Confusion table

	Predict 0	Predict 1
Actual 0	2	1
Actual 1	1	3

Want these to be high

1.3 Decision Boundary

$p = 2$



Predict $\hat{y} = 1$

Predict $\hat{y} = 2$

$$\hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 = 0$$

$$x_2 = -\frac{\hat{\beta}_0 + \hat{\beta}_1 x_1}{\hat{\beta}_2}$$

1.4 Multinomial Regression

Y has K classes. ($K > 2$). Pick a baseline class (class K).

$K = 4$

$k = 1, 2, \text{or } 3$

$$\log \frac{P(Y=k \mid X=x)}{P(Y=K \mid X=x)} = \beta_{k0} + x^T \beta_k$$

Each class gets its own β_{0k}, β_k

$$P(Y=k \mid X=x) = \frac{e^{\beta_{k0} + x^T \beta_k}}{1 + \sum_{l=1}^{K-1} e^{(\beta_{l0} + x^T \beta_l)}}$$

$$P(Y=K \mid X=x) = \frac{1}{1 + \sum_{l=1}^{K-1} e^{(\beta_{l0} + x^T \beta_l)}}$$

Softmax:

each class k gets one

$$p_k(x) = \frac{e^{\beta_{k0} + x^T \beta_k}}{\sum_{\ell=1}^{K} e^{\beta_{\ell 0} + x^T \beta_\ell}} \Rightarrow \sum_{k=1}^{K} p_k(x) = 1$$

sum across all classes

Prediction: predict class k with highest $p_k(x)$

Note: ridge & lasso works on logistic & multinomial regression. Add penalty ($\lambda \sum_j \beta_j^2$ or $\lambda \sum_j |\beta_j|$) to the negative log-likelihood.

Don't include β_{0k}

2. Generative vs. Discriminative

Logistic (Discriminative): $P(Y = k \mid X = x)$

↑ we don't use distribution of X or (X, Y)

Generative: $P(X, Y)$

$P(X \mid Y = k)$

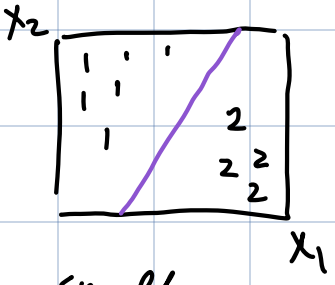
$P(Y = k)$

then use these, using Bayes' Theorem, to get

$$P(Y=k \mid X=x)$$

Why?

- Logistic fails if classes have perfect separation



- If $X \sim$ Normal, and n is small, generative is more accurate.

2.1 Bayes' Theorem

A, B

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B \mid A) P(A)}{P(B)} = \frac{P(A)\,P(B \mid A)}{P(B \cap A^c) + P(B \cap A)}$$

$$= \frac{P(A) P(B \mid A)}{P(A) P(B \mid A) + P(A^c) P(B \mid A^c)}$$

$$\underbrace{P(Y=k \mid X=x)}_{\text{posterior}} = \frac{P(Y=k)\,P(X=x \mid Y=k)}{\sum_{l=1}^{K} P(Y=l)\,P(X=x \mid Y=l)}$$

$$P(Y=k) = \pi_k \quad \text{prior for class } k$$

$$P(X=x \mid Y=k) = f_k(x) \quad \text{density of } x \text{ for class } k$$

posterior

$$p_k(x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}$$

$\pi_k f_k(x)$ — estimate these

$\sum_{l=1}^{K} \pi_l f_l(x)$ ← same for all k, ignore these

$$\text{So} \quad \hat{y}(x) = \underset{k}{\operatorname{argmax}} \; \pi_k f_k(x) = \underset{k}{\operatorname{argmax}} \left[\log \pi_k + \log f_k(x) \right]$$

$\hat{\pi}_k = \frac{n_k}{n}$ e.g. $\frac{\#\text{ of SPAM}}{\#\text{ of emails}}$

3. Linear Discriminant Analysis (LDA) Generative

3.1 Discriminant function

Assume $X \mid y = k \sim N(\mu_k, \Sigma)$ μ_k = mean of each predictor for class k

e.g. μ_{spam} = avg # of words

$p = 1$, $K = 2$ SPAM NONSPAM x_1 : # of words

